Optimizing distribution of drinking water in remote areas of Nepal

Biner Nicolas, Giacomelli Robin, Giora Triantafyllia, Margomenos Spyridon

April 2, 2026

Table of Contents

Problem Formulation

Goal: Distribute drinking water supply in remote areas of Nepal

Restrictions:

- All houses must have some minimum access to water
- Water amount must be proportional to their needs
- There are losses due to the terrain/pipe quality

Methods used:

- Lagrange Method
- Simulated Annealing

Problem Formulation

a : delivery-loss factor
b : the smallest required amount of water per household
d : demand per household
w : house priority/vulnerability factor
S : total available water supply

Continuous variables, smooth objective function, constrained

Lagrange Method

$\min_{x_1,\dots,x_n} \quad \sum_{i=1}^{n} w_i \, (d_i - x_i)^2$

subject to $\quad \sum_{i=1}^{n} a_i x_i = S$

$b_i \leq x_i \leq d_i, \quad i = 1, \dots, n$

Parameter restrictions: $\quad \mathrm{n} > 0, \quad \mathrm{d}_i \geq b_i, \quad S > 0, \quad a_i \geq 1, \quad w_i > 0$

$\mathrm{S}_{\min} = \sum_{i=1}^{n} a_i b_i, \qquad S_{\max} = \sum_{i=1}^{n} a_i d_i S_{\min} \leq S \leq S_{\max}$

Simulated Annealing

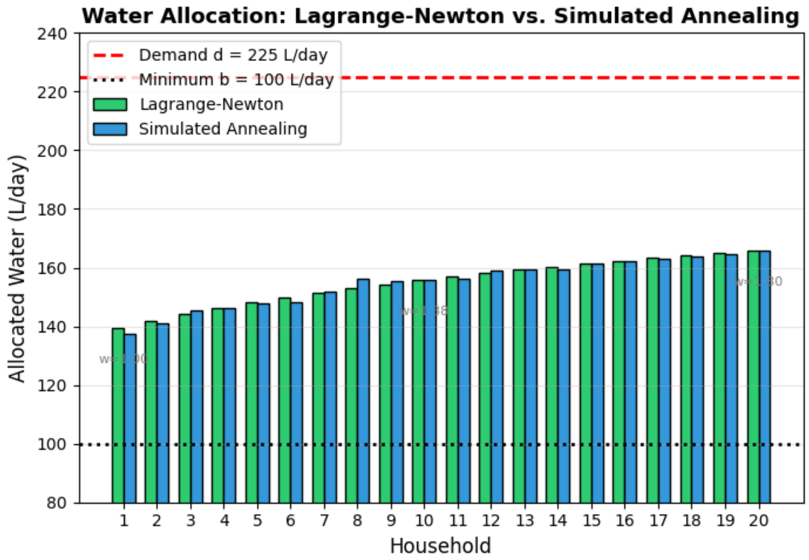
- $\mathrm{T_k} = T_0 \; * \left(\frac{1-k}{k\_max}\right)$ (with 50'000 Iterations)
  - Different Distributions
  - Prioritizing higher vulnerability

- $a_i * x_i = S$ -> remaining satisfied

- Exploration High T / low T

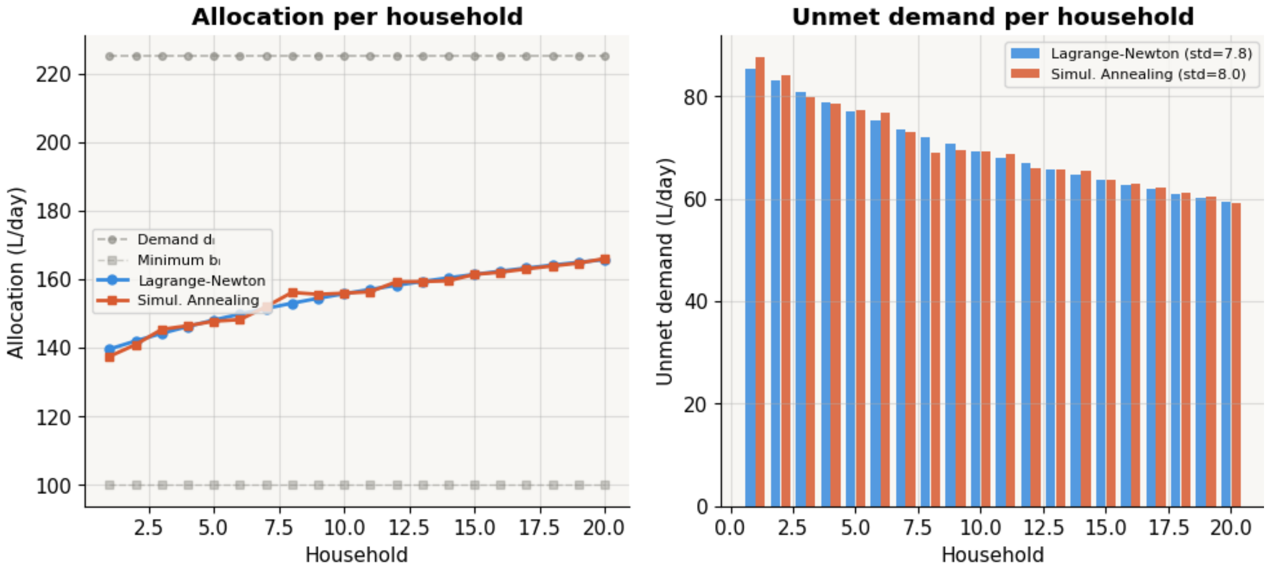
Method Comparison

Method Comparison: Lagrange-Newton vs Simulated Annealing



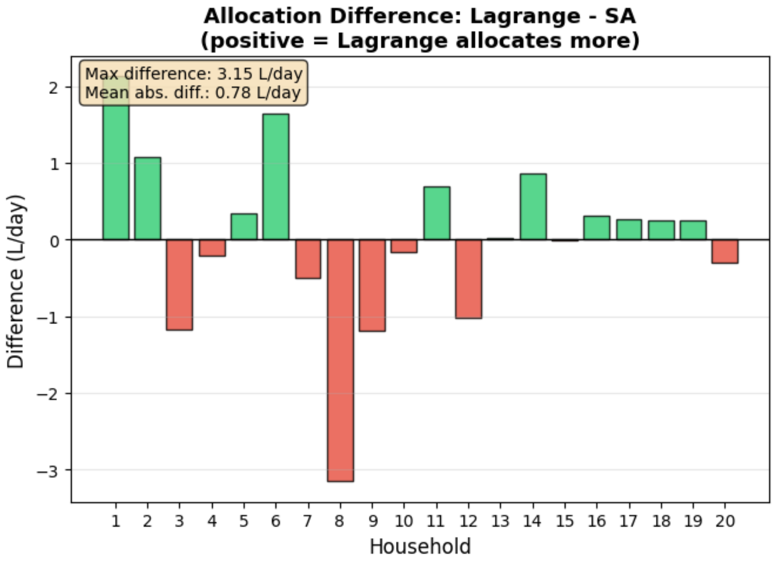
Method Comparison

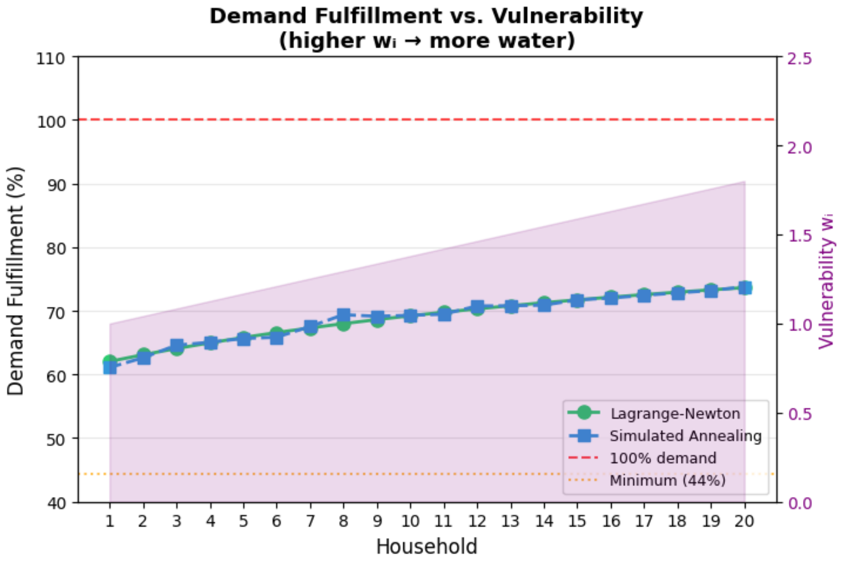
Method Comparison: Lagrange-Newton vs Simulated Annealing



Method Comparison

Method Comparison: Lagrange-Newton vs Simulated Annealing





Questions

Thank you